

1. Project and Package Creation

Purpose:

This step creates a container for the ETL workflow where all data flow components will reside.

Step Description:

- I opened Visual Studio 2026 Community and created a new Integration Services Project named "SSIS_Transformation_Lab"
- Visual Studio automatically created a default SSIS package called Package.dtsx that I later renamed to "CustomerETL.dtsx" where I will design my ETL process.

2. Adding a Data Flow Task

Purpose:

The Data Flow Task is the main container for moving data from source to destination while applying transformations.

Step Description:

- In the Control Flow tab, I dragged a Data Flow Task from the SSIS Toolbox into the design area.
- I renamed it to "Customer Data"

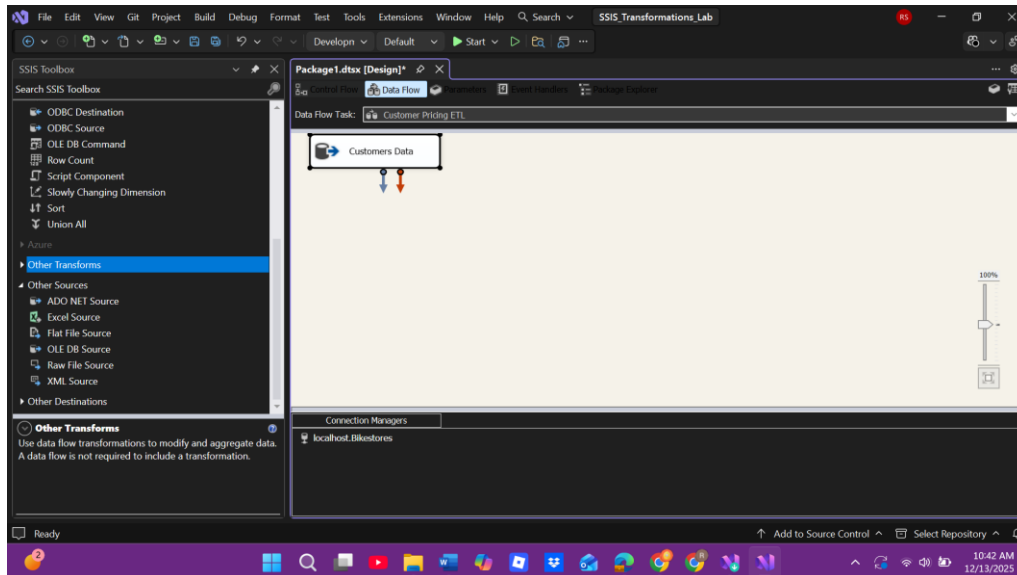
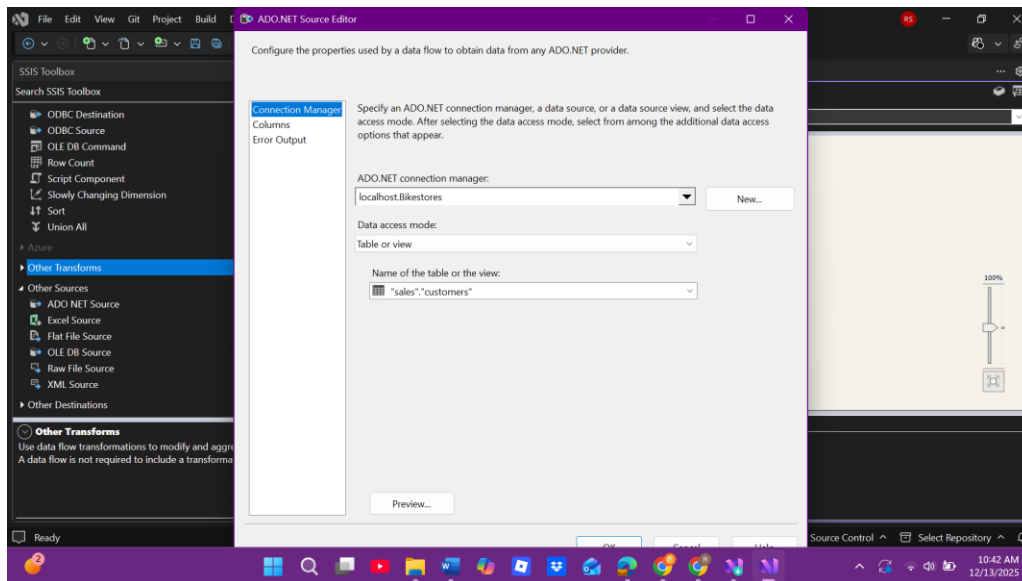
3. Adding the Source (ADO NET Source)

Purpose:

This step reads raw customer data from the database and passes it to the next transformation.

Step Description:

- In the Data Flow tab, I dragged ADO NET Source into the design area.
- Configured it to connect to my SQL Server database BikeStores.
- Selected the Sales.Customer table as the source.



4. Derived Column Transformation

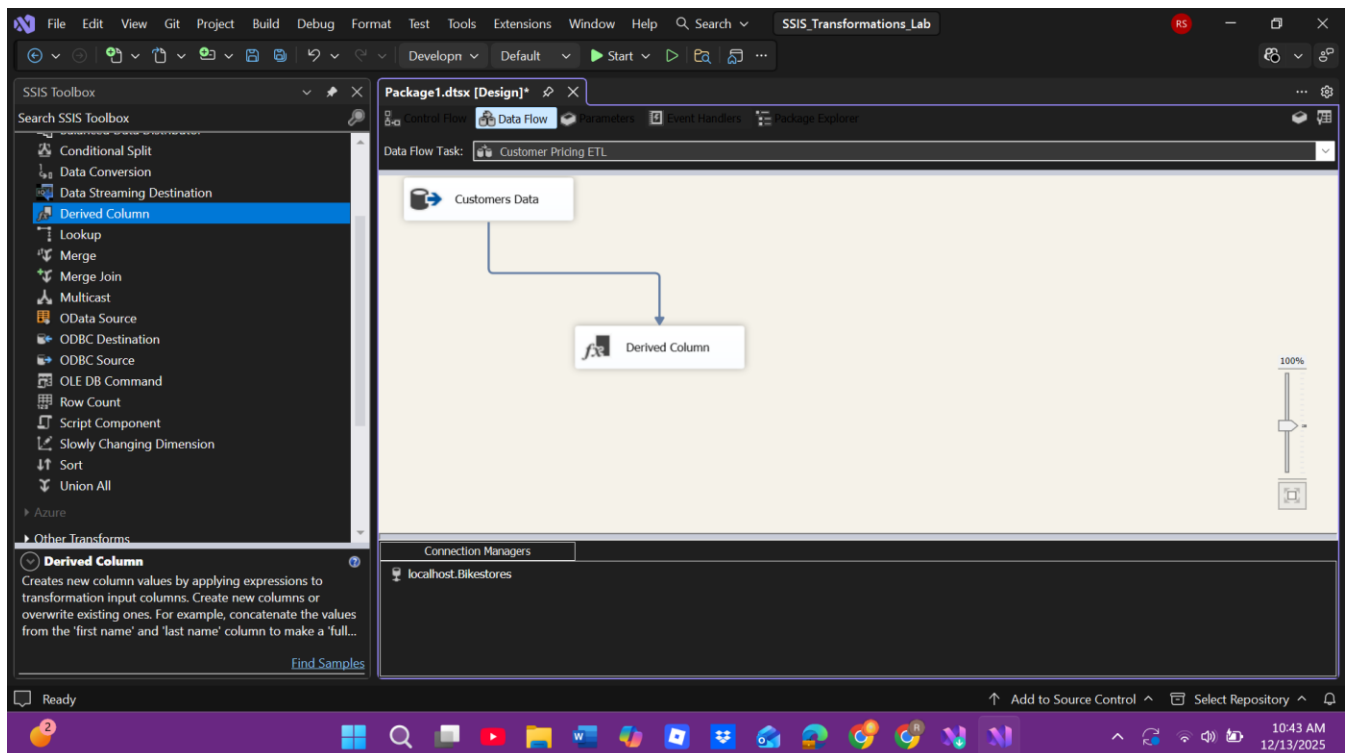
The Derived Column transformation allows you to create new columns or modify existing columns by applying expressions or calculations. It is used to shape and manipulate data as it flows through the ETL process.

Purpose:

The Derived Column allows creating new calculated columns or modifying existing ones. In this case, it combines first and last names.

Step Description:

- Dragged Derived Column into the Data Flow area.
- Connected the ADO NET Source to the Derived Column using the blue arrow.
- Configured a new column “FullName” as (first_name + " " + last_name)



5. Conditional Split Transformation

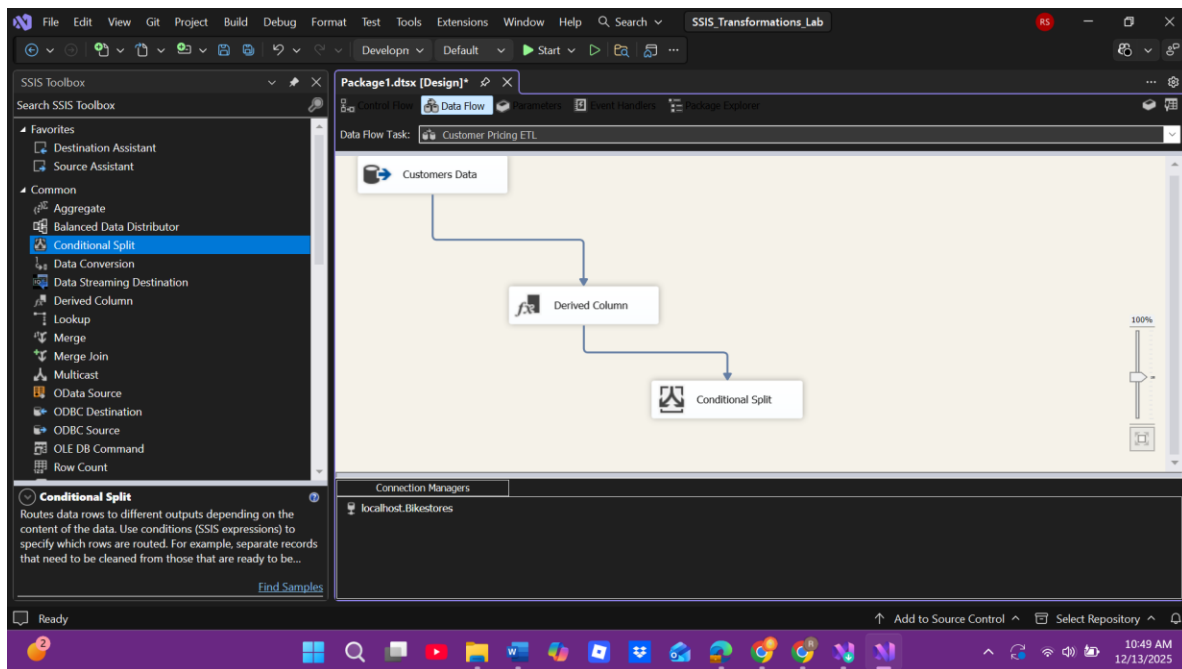
The Conditional Split transformation directs data rows to different outputs based on defined conditions. It is useful for categorizing or filtering data within the data flow according to specific rules.

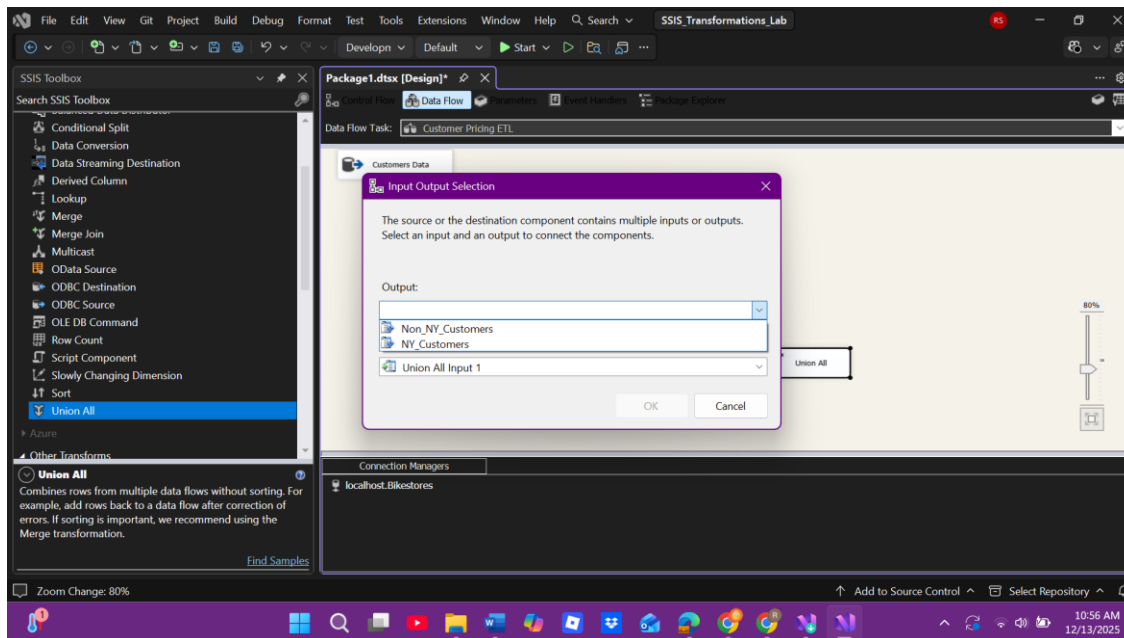
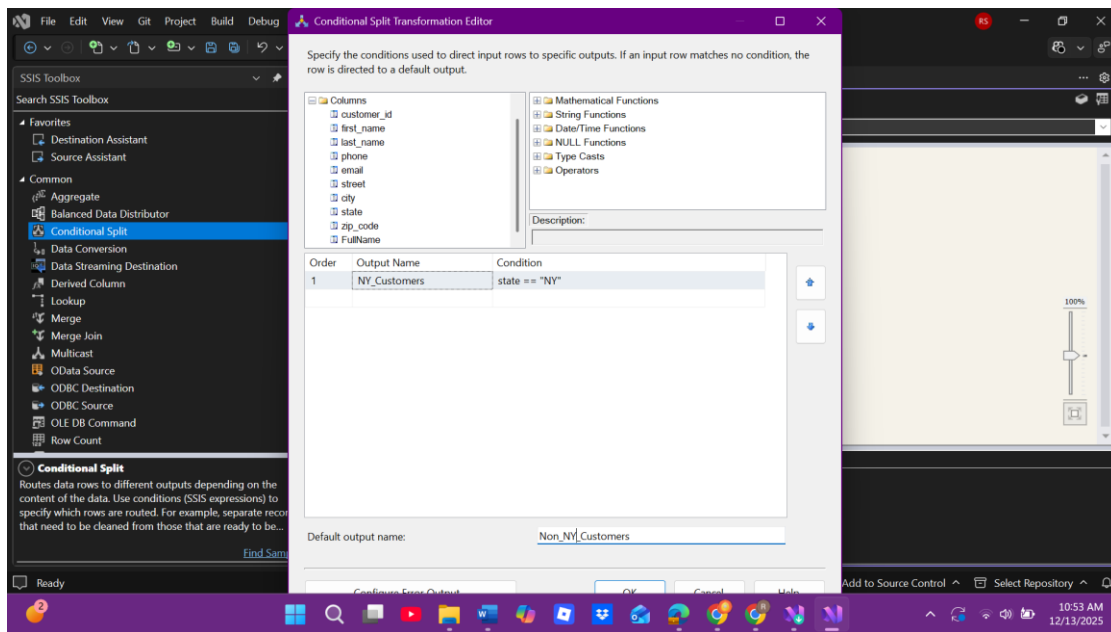
Purpose:

Conditional Split filters or routes data based on conditions. Useful for processing subsets differently.

Step Description:

- Dragged Conditional Split into the Data Flow area.
- Connected the Derived Column output to it.
- Created a condition state == "NY" to separate NY customers from others.





6. Data Conversion Transformation

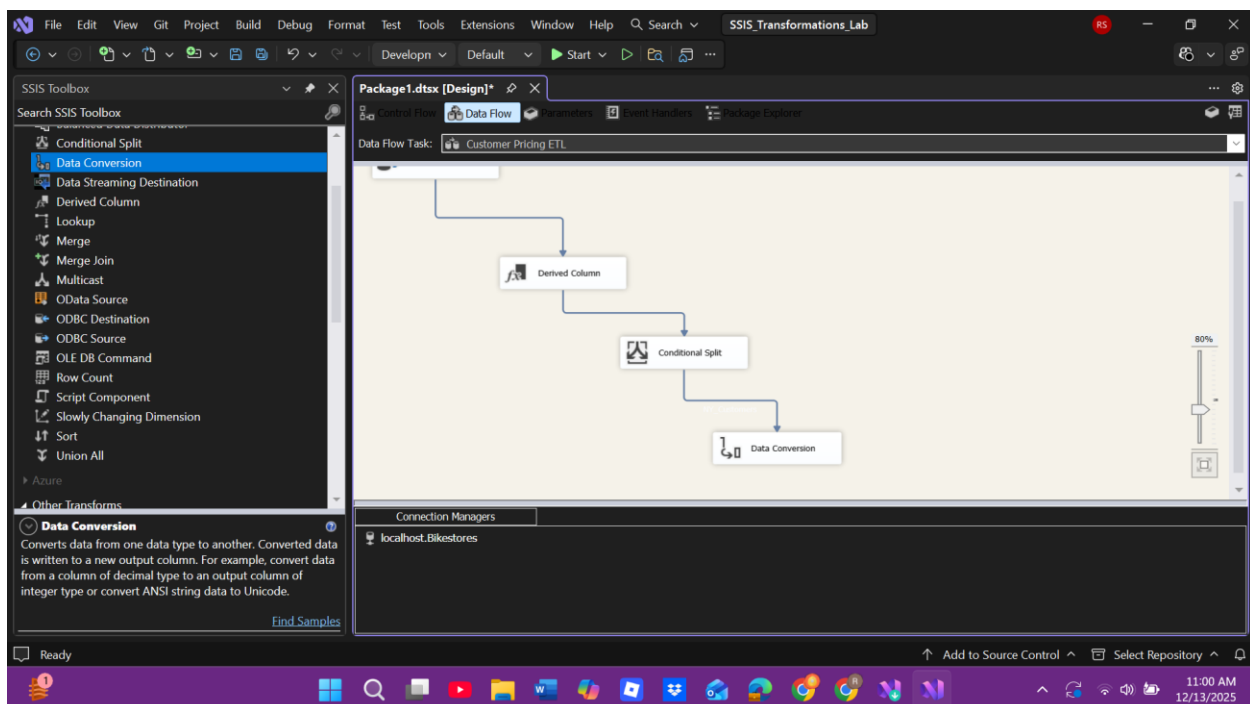
The Data Conversion transformation is used to change the data type of columns to match the requirements of the destination or other transformations. It ensures that the data flows correctly and is compatible with the target schema.

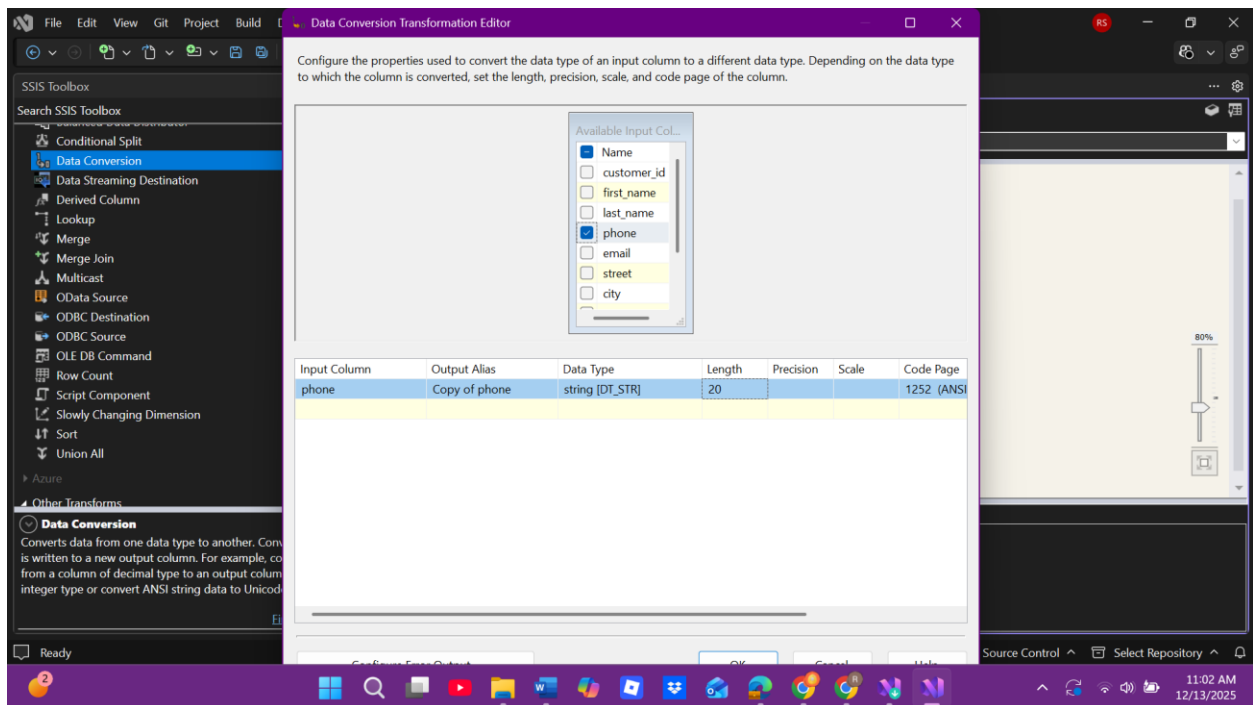
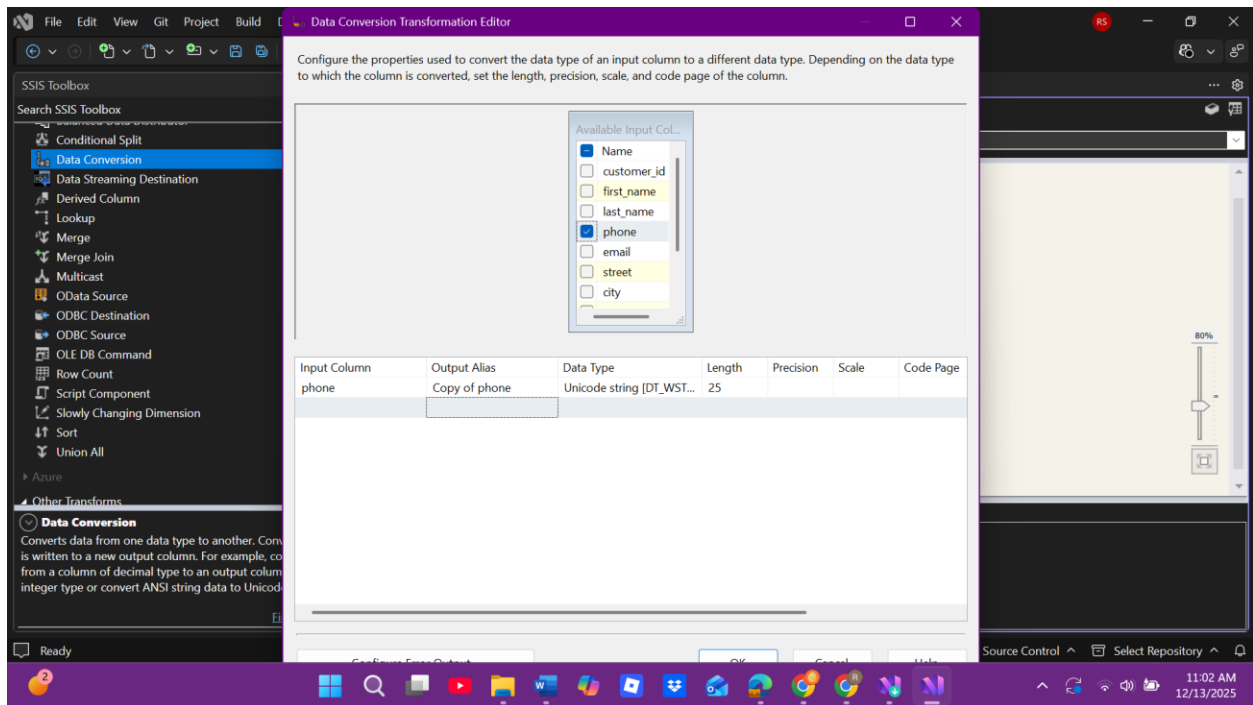
Purpose:

Data Conversion ensures that data types in the ETL flow match the destination schema. SSIS requires compatible types to avoid errors during insertion.

Step Description:

- After the Conditional Split, I dragged a Data Conversion transformation into the Data Flow area.
- Connected one of the outputs (NY_Customers) from the Conditional Split to the Data Conversion transformation using a blue arrow.
- Configured the Data Conversion to change data types and length of “phone”

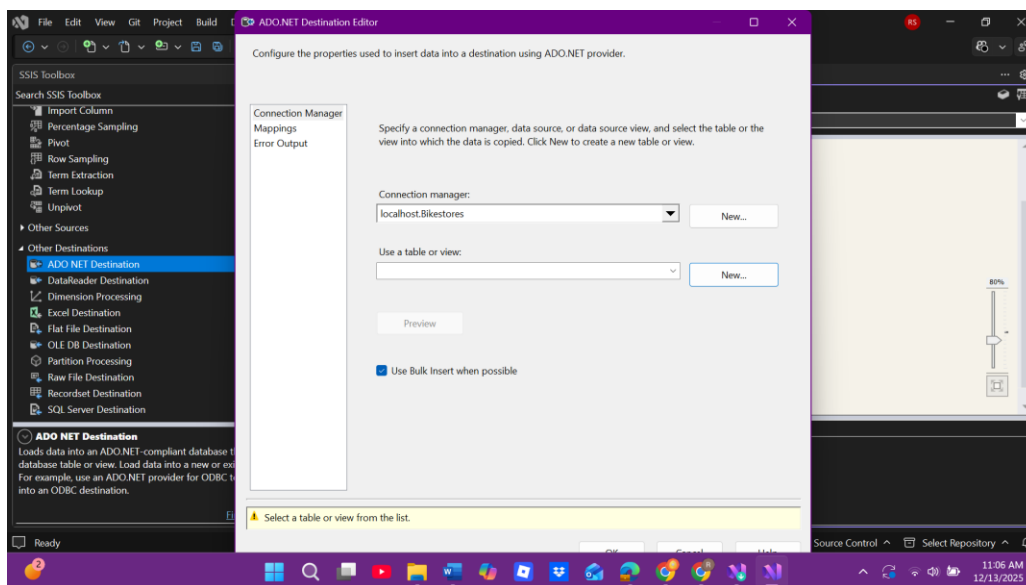
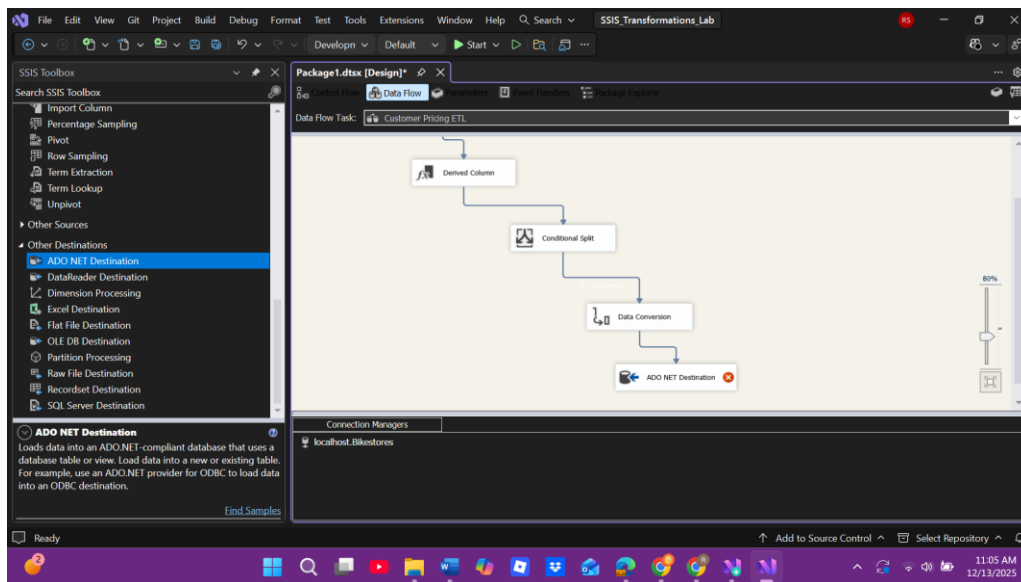


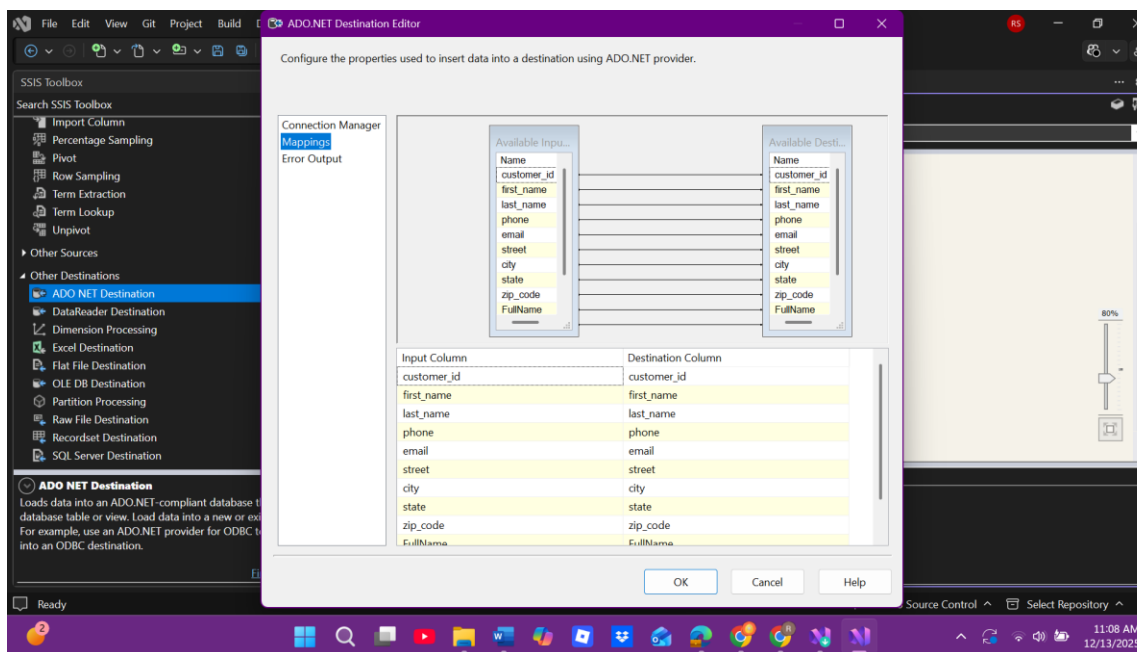
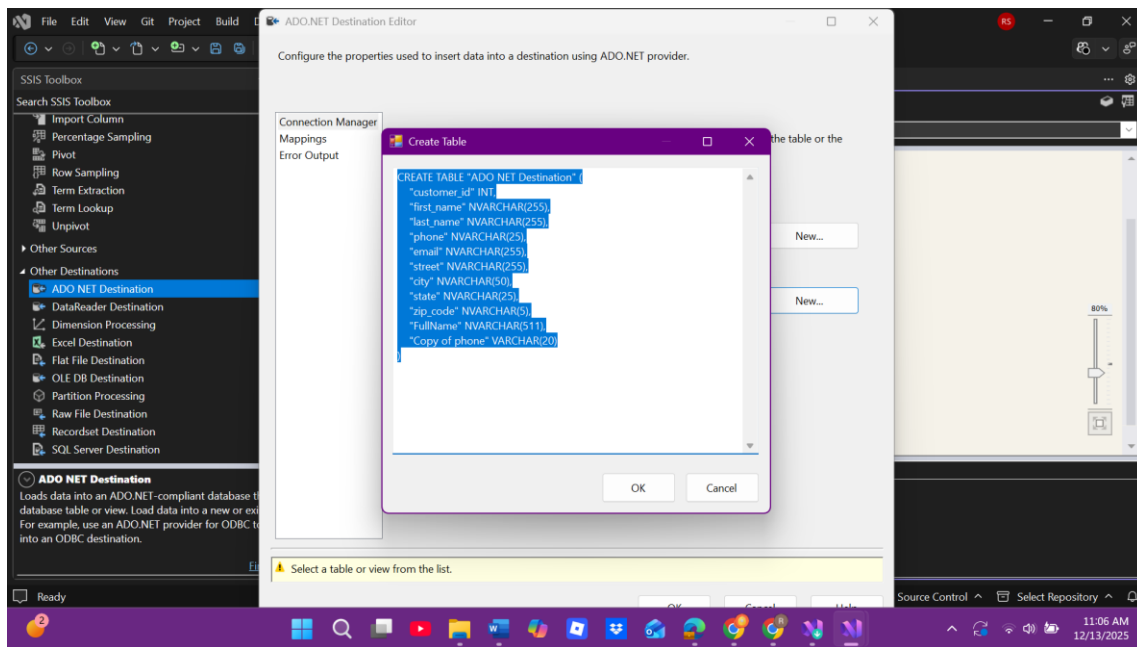


7. Destination (ADO NET Destination)

Step Description:

- Dragged ADO NET Destination into the Data Flow area.
- Connected the last transformation output to the destination.
- Configured it to insert data into “ADO NET Destinationn” table in the database.
- Mapped input columns to destination columns.

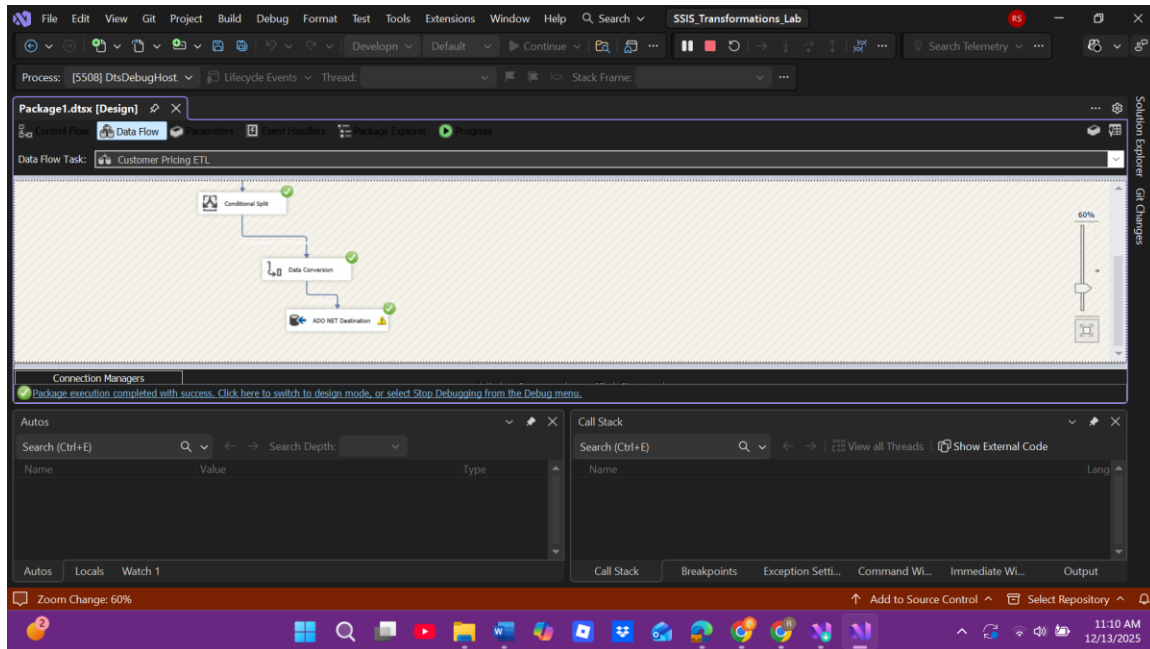
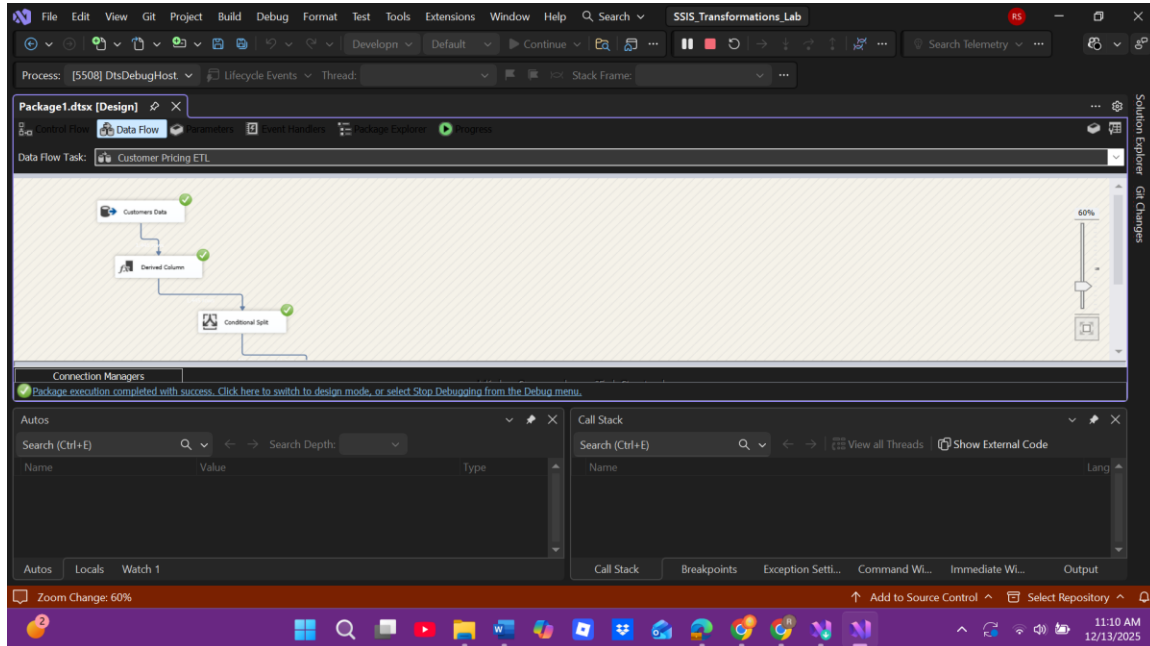


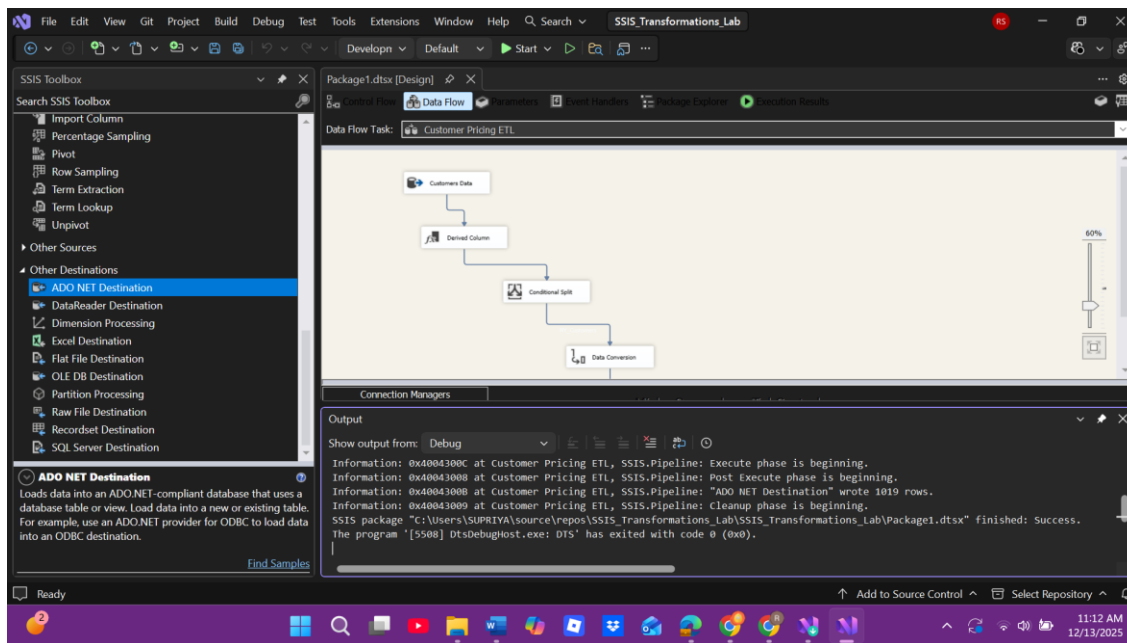


8. Testing the Package

Step Description:

- Clicked Start in Visual Studio to execute the package.
- Monitored the progress tab to confirm all tasks completed successfully.





9. Sample Output

Sample data confirms the Derived Column, Conditional Split, and Data Conversion transformations worked as intended.

Preview Query Results

Query result (up to the first 200 rows):

	custo...	first_n...	last_n...	phone	email	street	city	state	zip_code	FullNa...	Copy ...
1		Debra	Burke	NULL	debra...	9273 ...	Orcha...	NY	14127	Debra...	NULL
4		Daryl	Spence	NULL	daryl...	988 P...	Union...	NY	11553	Daryl ...	NULL
7		Lyndsey	Bean	NULL	lynds...	769 ...	Fairport	NY	14450	Lynds...	NULL
7		Latasha	Hays	(716) ...	latash...	7014 ...	Buffalo	NY	14215	Latas...	(716) ...
8		Jacqui...	Duncan	NULL	Jacqui...	15 Bl...	Jacks...	NY	11372	Jacqui...	NULL
9		Geno...	Baldwin	NULL	genov...	8550 ...	Port ...	NY	11050	Geno...	NULL
10		Pamella	New...	NULL	pamel...	476 C...	Monroe	NY	10950	Pamel...	NULL
11		Deshia...	Mend...	NULL	deshia...	8790 ...	Monroe	NY	10952	Deshia...	NULL
12		Robby	Sykes	(516) ...	robby...	486 R...	Hemp...	NY	11550	Robb...	(516) ...
15		Linnie	Branch	NULL	linnie...	314 S...	Platts...	NY	12901	Linnie...	NULL
16		Emmitt	Sanchez	(212) ...	emmi...	461 S...	New ...	NY	10002	Emmi...	(212) ...
17		Caren	Steph...	NULL	caren...	914 B...	Scars...	NY	10583	Caren...	NULL
18		Georg...	Hardin	NULL	georg...	474 C...	Canan...	NY	14424	Georg...	NULL
19		Lizette	Stein	NULL	lizett...	19 Gr...	Orcha...	NY	14127	Lizett...	NULL
22		Adelle	Larsen	NULL	adelle...	683 ...	East ...	NY	11731	Adelle...	NULL
23		Kaylee	English	NULL	kaylee...	8786 ...	Hollis	NY	11423	Kayle...	NULL
25		Regenia	Vaug...	NULL	regen...	44 S...	Maho...	NY	10541	Regen...	NULL
26		Theo	Riese	(562) ...	theo...	8755 ...	Long ...	NY	11561	Theo ...	(562) ...
27		Santos	Valencia	NULL	santo...	7479 ...	Sunny...	NY	11104	Santo...	NULL
28		Jeanice	Frost	NULL	jeanic...	76 De...	Ossini...	NY	10562	Jeanic...	NULL
29		Syreeta	Hend...	NULL	syreet...	193 S...	Maho...	NY	10541	Syreet...	NULL
34		Brittney	Wood...	NULL	brittn...	960 R...	East ...	NY	11731	Brittn...	NULL
35		Gullie...	Noble	NULL	guille...	6 Del...	Baldw...	NY	13027	Gullie...	NULL
36		Bernita	Mcd...	NULL	bernit...	2 Peg...	Uverp...	NY	13090	Bernit...	NULL
37		Melia	Brady	NULL	melia...	907 S...	Maspeth	NY	11378	Melia ...	NULL
38		Zelma	Brow...	NULL	zelma...	296 S...	Astoria	NY	11102	Zelma...	NULL
39		Janetta	Aguirre	(717) ...	janett...	214 S...	Lanca...	NY	14086	Janett...	(717) ...
41		Kathie	Freem...	NULL	kathie...	667 T...	Quee...	NY	12804	Kathie...	NULL
42		Tangela	Quinn	NULL	tangel...	4 S. P...	Richm...	NY	11418	Tange...	NULL
44		Onita	Johns	NULL	onita...	32 Gl...	Elmont	NY	11003	Onita ...	NULL
45		Bennett	Armst...	NULL	benne...	688 ...	Bethp...	NY	11714	Benne...	NULL

Close